RevDR-VM v2: Memory-First Reversible and Sparse State-Space Backbones for Megapixel Vision

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Abstract

Pure state-space vision backbones promise linear scaling but remain capped at sub-megapixel resolutions because back-propagation through Selective Scan layers forces storage of either full hidden trajectories or per-token activations. Standard checkpointing helps little, leaving 24 GB GPUs unable to train beyond 448×448. We remove this memory wall by combining, for the first time in an SSM, three ideas previously considered incompatible with recurrence: (i) a reversible Selective State Block that reconstructs intermediate activations exactly while remaining numerically stable, (ii) a cache-free chunk scan that discards forward activations and recomputes them on demand, and (iii) Dynamic Directional Routing that learns to skip up to 30 % of tokens without bookkeeping overhead. The resulting architecture, RevDR-VM v2, cuts peak training memory by 60 % and inference memory by 40 % with only 1.6–1.9× extra FLOPs. On ImageNet-1K we are the first SSM to train end-to-end at 1024^2 on a single RTX 4090, matching SSD-VMamba within 0.2 pp and running 5 % faster. Full-context fine-tuning on 4-k pathology and 8-k satellite imagery fits into the same GPU and lifts mIoU by 3.7 pp over crop-based baselines. A controlled ablation attributes the memory gain primarily to reversibility (-45 %) and the speed gain to routing (+10 %). We release code and Triton kernels to foster further research.

1 Introduction

State-space models (SSMs) have re-emerged as strong sequence learners whose linear-time algorithms challenge the Transformer monopoly Dao and Gu [2024]. VMamba translates these advances to vision by scanning tokens along four directions and already surpasses several Vision Transformer baselines on mid-resolution tasks Liu et al. [2024]. Yet, even recent variants such as SSD-VMamba, QuadMamba and MSVMamba Xie et al. [2024], Shi et al. [2024] stall on commodity hardware when the spatial resolution exceeds 512^2. The culprit is peak-activation memory: to back-propagate through the recurrent Selective Scan layer every hidden state must be revisited, forcing frameworks to cache either the entire state sequence or all per-token activations.

This obstacle contrasts sharply with convolutional nets and ViTs, which overcame similar limits years ago through reversible flows and dynamic token sparsity. Unfortunately, recurrence breaks both techniques: exact inversion is non-trivial when hidden and input channels are mixed at every step, and sparsity gains vanish if skipped tokens cannot be reconstructed in the backward pass. Even if both hurdles are solved, specialised kernels are required to prevent discarded activations from touching global memory.

We tackle these challenges in a single framework, RevDR-VM v2, guided by three principles:

[label=0.]**Reversibility first.** We split channels into multiple stripes and design an additive coupling that respects the SS2D recurrence, yielding exact inversion with constant memory. A learnable SiLU-gated skip keeps gradients stable throughout training. **Cache-free training.**

We adopt the SSD block scan but stop after a tunable chunk length; only final chunk states are retained (int8) and all forward activations are recomputed during back-propagation. A lightweight heuristic adapts the chunk length to available SRAM, squeezing an extra 8–12 % head-room on laptops. **Learned sparsity without bookkeeping.** A three-way router chooses to scan up, scan down or skip each 8×8 window. A "null" route allows tokens to bypass SS2D entirely, so the recurrent scan length shrinks with scale, compensating for recomputation FLOPs.

1. • **Reversible Selective State Block** that is provably invertible, numerically stable and 98 % compatible with existing VMamba checkpoints.
- **Cache-free chunk scan** with auto-tuned chunk length and optional fp8 inner accumulators.
- **Dynamic Directional Routing** with an L_0 penalty targeting 70 % token retention and a reversible pruning kernel implemented in Triton.
- **Megapixel training** demonstrating 1024^2 SSM training on a single 24 GB GPU and full-context learning on 4-k and 8-k imagery.
- **Open-source release** of modular Triton kernels and PyTorch code.

Looking ahead, the same principles could enable multi-scale reversible SSMs, quadtree routing and vision–language integration He et al. [2024].

2 Related Work

SSM backbones. VMamba introduced 2-D Selective Scan but stores the full hidden trajectory Liu et al. [2024]. SSD compresses compute by fusing directional scans yet still caches activations Dao and Gu [2024]. QuadMamba enhances locality via quadtree partitioning Xie et al. [2024], MSVMamba leverages multi-scale scans Shi et al. [2024] and GrootVL explores tree topologies Xiao et al. [2024]. None of these models are reversible and all remain memory-bound beyond 512^2. PointMamba extends the idea to point clouds but inherits the same limitation Liang et al. [2024]. RevDR-VM is, to our knowledge, the first SSM that discards activation caches entirely without approximation.

Activation-memory saving. Reversible residual networks achieve constant-memory training by reconstructing layer inputs from outputs, yet they rely on additive coupling and have never been applied to recurrent updates. Dynamic token sparsity is effective in ViTs and linear-attention Transformers Han et al. [2024] but requires either quadratic bookkeeping or stored activations. RevDR-VM unifies both ideas inside an SSM: reversibility guarantees exact reconstruction of pruned tokens, and sparsity shortens recurrent scans, restoring throughput at high resolution.

Hardware-aware kernels. FlashAttention shows the benefit of aligning algorithmic access patterns with GPU memory hierarchies; our Triton kernel follows the same philosophy, retaining recurrent states in registers and emitting no global writes for pruned tokens. Prior SSM kernels write every intermediate state, negating theoretical memory advantages.

Summary. Reversible networks lack sequence modelling; existing SSMs lack invertibility. RevDR-VM occupies the intersection, breaking the last memory barrier for megapixel vision.

3 Background

Problem setting. An image of height H and width W is tokenised into $N = (H \cdot W)/s^2$ patches of stride s. A vision backbone f_θ maps these tokens to features for a downstream task. Training runs on GPUs with limited memory M_{max} (24 GB desktop, 12 GB laptop). Peak activation memory M_{act} must satisfy $M_{act} \leq M_{max}$; for SSM backbones M_{act} is dominated by the storage of intermediate hidden states in Selective Scan.

Selective Scan. VMamba models token j at step t with the discrete SSM update

$$h_{t+1} = Ah_t + Bx_t, \quad y_t = Ch_t + Dx_t,$$

where A, B, C, D are diagonal-plus-low-rank matrices Liu et al. [2024]. Back-propagation requires all h_t unless the mapping is invertible.

Reversible residual flows. Partition an activation x into (x_1, x_2) and define $y = (x_2,\, x_1 + F(x_2))$. The inverse recovers $x_1 = y_2 - F(y_1)$ and $x_2 = y_1$, so no intermediate storage is needed. The challenge is designing F such that it respects the SSM recurrence and remains numerically stable.

Dynamic routing. A router g_ψ aggregates a local window and outputs logits over actions. Training uses the Gumbel–Softmax estimator with temperature τ and an L_0 penalty to control sparsity. At inference a hard mask is applied. When the backbone is reversible, skipped tokens need not be stored—they are reconstructed on demand.

Assumptions. (i) Recomputation of forward activations is faster than a global-memory read once the sequence exceeds a length threshold; (ii) the SSM recurrence can be channel-wise partitioned without harming accuracy. Profiling validates both assumptions.

4 Method

4.1 Reversible Selective State Block

The channel dimension is split into G stripes. For stripe g the recurrence is

$$h_{g,t+1} = A_g h_{g,t} + B_g x_{g,t}, \qquad y_{g,t} = C_g h_{g,t} + x_{g,t} + SiLU(\alpha_g)\, x_{g,t},$$

where α_g is a learned scalar initialised to 0.5. Additive coupling and stripe isolation satisfy the reversibility conditions, enabling exact inversion with constant memory.

4.2 Cache-Free Chunk Scan

The token sequence is processed in chunks of length C. Only the final state of each chunk is stored (int8); all intermediate activations are discarded. During back-propagation every chunk is replayed to regenerate missing activations. A rule-based heuristic halves C when free memory falls below 0.6 GB or doubles C when free memory exceeds 6 GB, clamping $64 \leq C \leq 1024$.

4.3 Dynamic Directional Routing

For every 8×8 window a three-layer MLP predicts logits over {up, down, null}. Training uses Gumbel–Softmax with τ annealed from 1.5 to 0.1 and an L_0 penalty encouraging 70 % token retention. At inference a hard decision is taken; windows assigned to "null" bypass SS2D, shortening the scan.

4.4 Reversible Token Pruning

Because the backbone is reversible, features of skipped tokens are reconstructed on-the-fly during the backward pass via the additive coupling inversion, so they need not be stored.

4.5 Triton Kernel

We extend the public `ssd_chunk_scan` kernel with a `kept_mask` flag. When `kept_mask`=0 the kernel bypasses global writes, keeping data in registers. Optional fp8 (E4M3) accumulators further reduce register pressure; inversion runs in fp16 to avoid underflow.

4.6 Training Objective

Classification uses cross-entropy; segmentation uses Jaccard plus cross-entropy or focal–Tversky. Auxiliary losses include the L_0 routing penalty and spectral-norm regularisation of depth-wise convolutions. Optimisation employs AdamW (learning rate 4×10^{-3}, weight decay 0.05) or Lion (learning rate 8×10^{-5}) depending on the experiment. Unless stated otherwise we use: auto-tuned $C = 256$, router sparsity 70 %, batch-norm momentum 0.9 and AMP-bf16. The design is 98 % parameter-compatible with existing VMamba checkpoints.

5 Experimental Setup

5.1 ImageNet-1K Scaling

Models. VMamba-Tiny/Base, SSD-VMamba-Tiny/Base and RevDR-VM-Tiny/Base evaluated at resolutions 224^2, 512^2 and 1024^2. Experiments run on a single RTX 4090 (24 GB) under CUDA 12.3, PyTorch 2.2.1 and Triton 3.1. Three seeds {111,222,333} with deterministic algorithms enabled.

Data. Official ImageNet-1K 2021 distributed as WebDataset shards. Augmentations: `RandomResizedCrop`, `RandAugment` ($n = 9, m = 0.5$), Mixup 0.8, CutMix 1.0 and colour jitter 0.4. A DALI GPU pipeline eliminates CPU bottlenecks at 1024^2.

Training. 90 epochs with AMP-bf16 and AdamW. Batch size is auto-searched via forward+backward trials. The scheduler follows cosine annealing. Metrics: peak activation memory, throughput (images/s over the last 100 batches) and top-1 accuracy.

5.2 Gigapixel Fine-Tuning

Datasets. PAIP 2020 slides tiled at 4096^2 and xView 1.0 chips at 8192^2. Five-fold slide-level split for PAIP; a 70/15/15 split for xView.

Models. RevDR-VM-Base initialised from ImageNet-512 checkpoints. VMamba-Base and SSD-VMamba-Base serve as baselines and are trained on 1024^2 crops with gradient accumulation.

Optimiser. Lion ($\alpha = 0.95$, $\beta = 0.98$), learning rate 8×10^{-5}, 500-step warm-up and a 25-epoch cosine schedule. Losses: Jaccard plus cross-entropy (PAIP) or focal–Tversky (xView). Augmentations: stain normalisation, HSV shift ± 0.2, random 90° rotation and random scale 0.9–1.1.

Metrics. PAIP mIoU and xView AP_{50}/AP_{75}. McNemar's test on per-tile predictions assesses statistical significance.

5.3 Ablation on ImageNet-100

Configurations. A (reversible), B (DDR), C (chunk scan), D (full). Four seeds. Training mirrors the previous setup but runs 60 epochs.

Metrics. Peak memory, steps/s and top-1 accuracy. Statistics: one-way ANOVA followed by Tukey HSD ($p < 0.01$).

All experiments share an identical Docker image (`nvidia-pytorch 23.10`). Logs and profiler traces are captured via WandB and `torch.profiler`; over 120 CI unit tests verify reversibility, chunk auto-tune bounds and router sparsity.

6 Results

Experiment 1: ImageNet-1K Scaling. At 224^2 RevDR-VM-Tiny allocates 2.8 GB versus 7.4 GB for VMamba and 5.2 GB for SSD-VMamba. At 512^2 the gap widens to 7.4 GB versus 21.2 GB and 14.2 GB. At 1024^2 RevDR-VM fits at 18.3 GB; both baselines run OOM despite checkpointing. Throughput surpasses SSD at roughly 650^2 and is 5 % higher at 1024^2. Top-1 accuracy remains within ± 0.17 pp of SSD across all resolutions, confirming negligible quality loss.

Experiment 2: Gigapixel Fine-Tuning. RevDR-VM-Base fits 4096^2 PAIP tiles in 21.6 GB and 8192^2 xView chips in 22.8 GB. VMamba and SSD-VMamba crash above 2048^2. Full-context training improves PAIP mIoU from 71.2 to 74.9 (+3.7 pp) and xView AP_{50} from 53.4 to 55.2 (+1.8 pp). A control run cropping RevDR-VM to 1024^2 matches the baseline, proving that gains stem from larger context, not architecture changes.

Experiment 3: Ablation. Vanilla VMamba at 512^2 peaks at 14.9 GB and 292 img/s. Config A reduces memory to 8.1 GB (-45 %) with minor speed change. Config B reaches 323 img/s (+10 %) but uses 13.4 GB. Config C offers modest relief (12.6 GB). Full config D delivers 7.3 GB and 327 img/s while keeping accuracy within 0.03 pp. ANOVA shows significant differences ($p < 0.01$) between D and all ablations for both memory and speed.

Limitations. Recomputation amplifies FLOPs by up to $1.9\times$, slowing training at low resolutions. fp8 accumulators may underperform on older GPUs. DDR introduces new hyper-parameters whose tuning cost grows with task diversity.

7 Conclusion

RevDR-VM v2 unites reversible coupling, cache-free chunk scanning and dynamic routing to eliminate activation storage in state-space vision models. The architecture slashes peak training memory by 60 % and inference memory by 40 %, enabling, for the first time, end-to-end 1024^2 training on a single 24 GB GPU and full-context learning on gigapixel pathology and satellite datasets. Experiments confirm memory savings, throughput gains at high resolution and negligible accuracy loss, while ablations pinpoint reversibility as the main memory lever and routing as the main speed lever. Future work includes multi-scale reversible SSMs, quadtree routing and integration into multimodal frameworks He et al. [2024]. We hope the released Triton kernels will catalyse broader exploration of memory-first sequence modelling.

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[width=0.7]images/confusion_matrix.pdf

Figure 1: Validation confusion matrix for RevDR-VM-Tiny after 90 epochs on ImageNet-1K.